Beyond Crest Factor: A Five-Corpus Audit of Signal Associations in Speech Anti-Spoofing

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Paper under active development for anonymous review

Abstract—Speech anti-spoofing systems can obtain convincing benchmark scores while responding to corpus-specific acoustic cues rather than synthesis artifacts. We distinguish three propositions: that a waveform feature separates labels, associates with a detector score within label, and changes a score under a content-preserving intervention. The locked audit registers 28 descriptors on three waveform views, eight published detector score artifacts, three discovery corpora, and two held-out confirmation corpora. Across 6,720 within-class screen cells, a discovery-frozen crest-factor candidate is negative for Spectra-AASIST in all three discovery corpora but attenuates sharply. In In-the-Wild its adjusted association is small ($\rho = -0.0326$), while in ASVspoof5 it is null after the registered controls ($\rho = 0.0014$), despite a negative unadjusted association. Thus crest factor fails the fixed five-corpus portability condition. Nineteen other pre-registered feature/view/class units meet the descriptive association rule, but are not promoted post hoc to paired interventions. The frozen crest/control panel also fails its 90% per-arm quality-retention gate before detector scoring. The result is a conservative negative finding about crest factor, together with a reproducible association atlas; it makes no causal shortcut claim.

Index Terms—speech deepfake detection, anti-spoofing, shortcut learning, causal audit, robustness

I. Introduction

Speech anti-spoofing countermeasures must distinguish bona fide speech from synthetic, converted, or replayed speech. Modern systems can learn effective spectro-temporal representations; for example, AASIST uses integrated spectro-temporal graph attention [1]. However, strong scores do not by themselves establish that a system responds to the intended manipulation artifacts. Müller et al. found that silence duration can correlate with the target label in ASVspoof data and that this cue alone can yield strong discrimination [2]. More generally, controlled shortcut conditions can produce near-perfect or worse-than-chance countermeasures without a reliable signal of the intended task competence [3]. Recent cross-domain evidence further indicates that audio-deepfake generalization gaps can reflect distributional difference, rather than only a harder version of the same task [4].

The present study begins with a practical observation: an interpretable feature—including crest factor—may be label-discriminative, correlated with a detector score, and causally influential under waveform transformation; these are different claims. Confusing them risks attributing model reliance to a feature that merely tracks label prevalence, corpus identity, duration, or loudness. We therefore audit published per-sample scores without recomputing baseline inference and use a staged, locked decision process.

This paper makes the following methodological contributions.

- It registers a cross-dataset, multi-model association design that separates label effects from within-class score-feature associations and fixes the acceptance criterion before confirmation data are analyzed.
- It specifies quality-gated paired waveform interventions that test causal score sensitivity rather than treating correlation as a shortcut.
- It defines a conditional mitigation experiment: a feature-conditioned head with transformation consistency is evaluated only as a consequence of audit evidence, against an equivalent training recipe and held-out corpora.

The manuscript is intentionally an initial, evidence-conservative draft. Sections III–IV distinguish locked methodology, three discovery screens, and pending confirmation. The title and narrative will be updated only after the documented outer-loop evidence gate.

II. Background and audit target

The ASVspoof 2019 challenge provides a common setting for logical-access spoofing research [5]. Let $y_i \in \{0, 1\}$ denote the registered label of utterance i, f_i an interpretable waveform feature, and $s_{m,i}$ the published score of model m. Each score artifact is oriented to increasing spoof evidence only after a label-direction check; artifacts that fail an exact sample-ID join or an orientation check are excluded rather than heuristically repaired. Let c_i contain registered nuisance variables such as duration, integrated loudness, and available speaker, attack, or codec metadata.

The first audit target is the within-class association $\rho_{m,y} = \text{Spearman}(f_i, s_{m,i} \mid y_i = y)$ and its rank-residual analogue after controlling for c_i. This target is not a causal estimand: a residual association may still be caused by unobserved corpus properties. A subsequent paired transformation supplies the causal target $\Delta s_{m,i} = s_m(T(f)_i) - s_m(x_i)$, subject to quality gates that limit changes to speech content and loudness. The distinction structures the three hypotheses below.

III. Locked study design

A. Inputs, feature registry, and provenance

All datasets, model revisions, and published score artifacts are pinned in the repository manifest. Discovery uses ASVspoof 2019 LA, ASVspoof 2021 LA, and ASVspoof 2021 DF; In-the-Wild and ASVspoof 5 are held out for confirmation. The 2021 challenge adds channel/compression variation and unmatched evaluation conditions, so these corpora are not treated as interchangeable replications of the 2019 setting [6]. The score panel consists of Spectra-AASIST, AASIST, Res2TCNGuard, RawTFNet, WhisperMFCCMesoNet, W2V2-AASIST, XLSR-SLS, and RawBMamba. Where a published score artifact is absent, the cell is reported as missing and is never imputed. This design uses the Arena score artifacts as published evidence, rather than spending additional compute to reproduce their baseline inference.

The frozen v1_28 registry extracts 28 amplitude, spectral, excitation, phase, and modulation descriptors from full-waveform, deterministic-crop, and pre-emphasized views. Voice-dependent descriptors retain missing values for unvoiced or unsuitable signals; they are not converted to zero. For corpora above 50,000 examples, the analysis selects a deterministic, class-balanced subset using seed 2609 and at most 5,000 examples per class.

B. H1: association and portability

For each model, dataset, class, feature, and waveform view, H1 reports feature label AUROC, within-class Spearman association, and partial rank association. P-values are adjusted with Benjamini–Hochberg correction. Candidate estimates receive 2,000 clustered bootstrap replicates, with a speaker or source utterance cluster when metadata permits. Exclusions are limited to missing stable IDs, unreadable waveforms, required labels, or published scores; every count is reported.

A feature is portable only if it has the same direction in at least four of the five core datasets, is significant in both held-out confirmation datasets, and is present in at least five of eight models. This criterion is fixed before confirmation analysis. Crest factor is one candidate in this registry, not the pre-supposed conclusion.

C. H2: quality-gated paired interventions

H2 is run only on H1-frozen candidates, with 500 examples per class per core dataset and the intervention panel Spectra-AASIST, AASIST, Res2TCNGuard, and W2V2-AASIST. The registered transformations comprise loudness-matched dynamic-range compression, leading/trailing silence standardization, spectral tilt, stable all-pass phase perturbation, and gain and polarity negative controls. A transformed example is retained only if STOI is at least 0.95, original-to-transformed transcript WER is at most 5%, loudness drift is at most 0.2 LU, and no transform-induced clipping occurs. An arm is dropped if fewer than 90% of its examples pass. Causal sensitivity requires a nonzero bootstrap interval, a standardized paired score change of at least 0.2, and consistent direction in at least three datasets and three of four models.

D. H3: conditional feature-conditioned mitigation

H3 is confirmatory only after an H2 causal result; otherwise it is explicitly exploratory. Using ASVspoof 2019 LA train/dev only for training and tuning, we compare an equivalent-recipe baseline head, logistic late score-feature fusion, a learned feature-conditioned head, and that head with consistency loss on H2-accepted transformations. Spectra-AASIST and AASIST are the first backbones. Each newly trained variant uses BF16, seeds 13/37/73, and an independently measured fastest batch/worker setting on one RTX 6000 Ada GPU. Success requires at least 10% relative macro held-out EER improvement on two or more datasets, a paired interval excluding zero, and no more than a 0.2-point in-domain EER degradation. This deliberately conservative use of frozen representations is motivated by recent generalization and calibration results, but it does not assume that feature conditioning itself is beneficial [7].

IV. Results

Table I records the completed H1 evidence, and Fig. 1 visualizes the registry-wide associations that satisfy the fixed rule. The aggregate is exhaustive over the registered matrix, but it is not a post-hoc intervention selector. No H2 delta, EER, or causal claim is reported.

V. Limitations and responsible reporting

The proposed audit cannot prove that every unmeasured cue is irrelevant, and its interventions cover only registered transformations that pass automated quality gates. Metadata availability differs by corpus, so the partial analyses must report which controls are present rather than implying uniform adjustment. Published score artifacts broaden architecture coverage but do not replace access to all model internals. The conditional mitigation can establish only a bounded robustness result under the registered splits and transformations, not universal safety against new generators or post-processing pipelines.

The H1 matrix now contains two held-out corpora, which expose why a repeated discovery direction alone is insufficient: the crest candidate is null after adjustment in ASVspoof5. The 19 registry-wide associations are descriptive screen results, so they neither identify a causal mechanism nor authorize post-hoc waveform transformations. The H2 crest run was frozen before those confirmation outcomes, but three of its four arms fail the 90% retention gate before detector scoring. It therefore supplies no paired-sensitivity estimate.

VI. Conclusion

We present a reproducible path from a score-feature correlation to a testable claim about detector sensitivity: exact artifact joins, within-class adjusted associations,

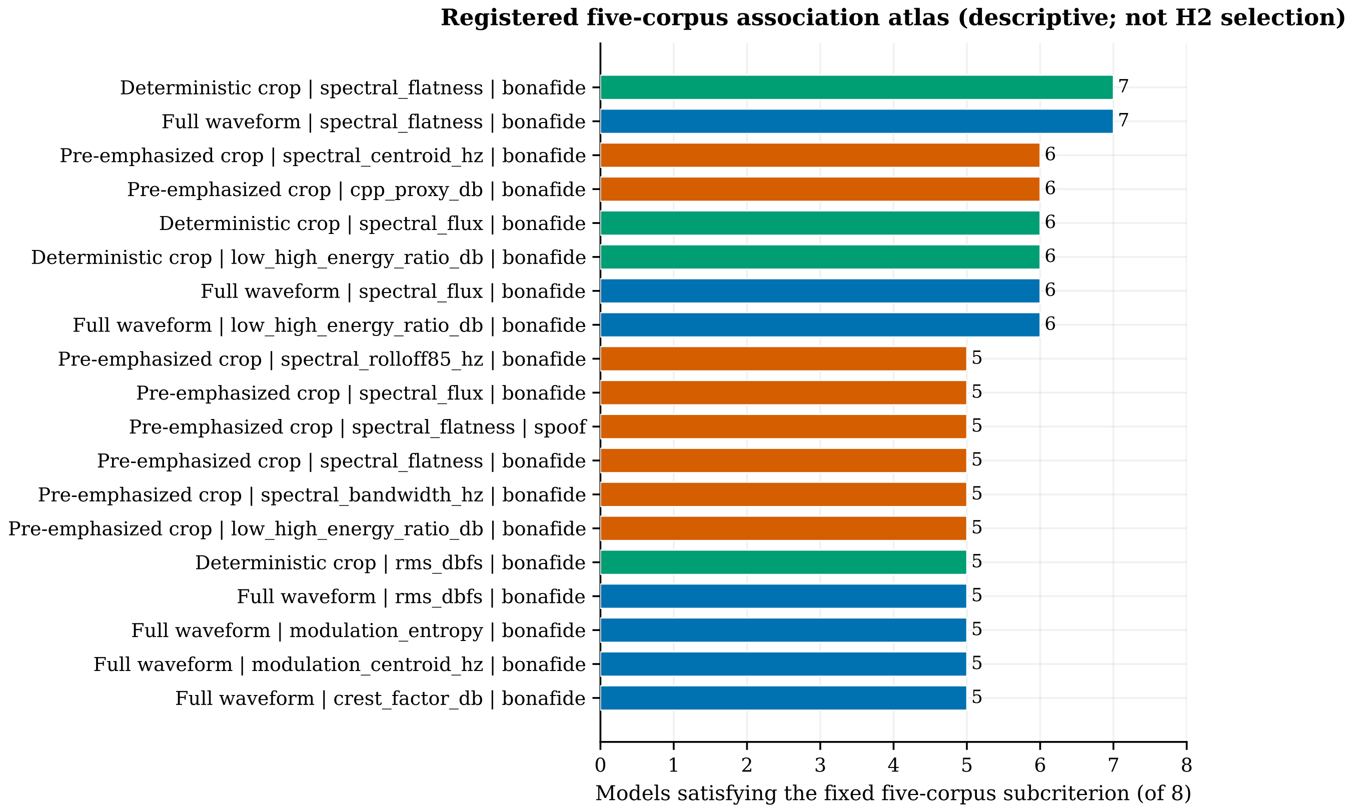

Fig. 1. Five-corpus association atlas. Each bar is a registered feature/view/class unit meeting the fixed partial-Spearman/BH rule in at least five of eight score artifacts; colors identify the waveform view. These 19 descriptive associations were not re-ranked or promoted to H2 after confirmation data were inspected. The frozen crest candidate is absent because it fails the same rule.

quality-gated paired interventions, and a conditional robustness test. Five corpora show that repeated crest-factor direction in discovery data does not survive the registered portability condition. The result also exposes a broader association atlas without converting it into post-hoc interventions. The pre-frozen crest transformations fail their own quality gate before score measurement. The paper therefore makes no causal feature-reliance claim.

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TABLE I
Completed H1 evidence. The five-corpus association outcome is not a causal result and does not select new H2 candidates.

Artifact	Observed status
Spectra-AASIST score/label join on pinned ASVspoof 2019 LA	71,237 scores join one-to-one; raw score is oriented toward bona fide evidence and is negated for the canonical spoof-evidence analysis.
Feature-registry pilot on deterministic ASVspoof 2019 LA sample	200 utterances $\times$ 3 waveform views; non-voice descriptors complete and voice-quality descriptors available for 95% of records.
H1 screens on three discovery plus two held-out corpora	6,720 within-class tests (5 corpora $\times$ 8 models $\times$ 2 classes $\times$ 3 views $\times$ 28 features), with exact joins for every model and corpus.
Fixed crest-factor slice: full waveform, spoof class	Spectra-AASIST partial ρ is -0.454, -0.099, and -0.065 on 2019 LA, 2021 LA, and 2021 DF. It is -0.0326 in In-the-Wild but 0.0014 in ASVspoof5 after registered adjustment; it therefore fails portability.
Registered five-corpus association rule	19/168 feature/view/class units meet the descriptive rule ($\geq 5/8$ models); none is added post hoc to H2.
Frozen H2 crest/control quality panel	DRC-3, DRC-6, and gain retain 18.1%, 0.6%, and 64.6%, respectively, below 90%; no detector scores, paired deltas, or H3 training are produced.